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Employment

***CITA National Fellow**, McGill University: 2024-2026*

***TSI Postdoctoral Fellow**, McGill University: 2024-2026*

Research Associate in Radio Cosmology, University of Manchester: 2022-2024

Postdoctoral Research Assistant, Queen Mary University of London: 2021-2022

Other Appointments

Visiting Postdoc, University of Cambridge, 2022 - 2023

Research Assistant, University of Washington, 2017-2021

ASTRO3D Science Visitor, Oct 2019

Teaching Assistant, University of Washington, 2015-2017

Education

PhD, Physics, University of Washington, 2015-2021

BS, Physics, University of California Los Angeles, 2011-2015

References

Prof. Adrian Liu: adrian.liu2@mcgill.ca

Dr. Phil Bull: phil.bull@manchester.ac.uk

Prof. Miguel Morales: miguelfm@uw.edu

Publications

... indicates trailing author

[1] Abdurashidova, Zuhra, Adams, Tyrone, ... **Wilensky**, **Michael J.**, et al. "First Results from HERA Phase II". In: *Astrophysical Journal* 998.1, 33 (Feb. 2026), p. 33. DOI: 10.3847/1538-4357/ae2d54. arXiv: 2511.21289 [astro-ph.CO].

[2] Ducharme, Jade M., Li, Andrei, **Wilensky, Michael J.**, et al. "Interference Meets Inference: Bayesian Time-Series Modeling of Radio-Frequency Interference". In: *arXiv e-prints*, arXiv:2609.26904 (Sept. 2026), arXiv:2609.26904. DOI: 10.48550/arXiv.2609.26904. arXiv: 2609.26904 [astro-ph.IM].

[3] Ducharme, Jade M., Pober, Jonathan C., and **Wilensky, Michael J.** "Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies". In: *arXiv e-prints*, arXiv:2609.35983 (Sept. 2026), arXiv:2609.35983. DOI: 10.48550/arXiv.2609.35983. arXiv: 2609.35983 [astro-ph.IM].

[4] Dutta, Sohini, Bull, Philip, ... **Wilensky, Michael J.**, et al. "Bayesian power spectrum estimation with modelling of systematic effects in delay-fringe rate space". In: *arXiv e-prints*, arXiv:2604.26885 (Apr. 2026), arXiv:2604.26885. DOI: 10.48550/arXiv.2604.26885. arXiv: 2604.26885 [astro-ph.CO].

[5] Hailemariam, Mekuanint K., Leeuw, Lerothodi L., ... **Wilensky**, **Michael J.**, et al. "Results from the investigation of temporal discontinuity in the Hydrogen Epoch of Reionization Array data". In: *RAS Techniques and Instruments* 5, rzag030 (Jan. 2026), rzag030. DOI: 10.1093/rasti/rzag030.

[6] **Wilensky, Michael J.** and Chokshi, Aman. "Inferring radio beams from sparse sampling patterns: a case study with the Murchison Widefield Array". In: *arXiv e-prints*, arXiv:2610.00466 (Sept. 2026), arXiv:2610.00466. DOI: 10.48550/arXiv.2610.00466. arXiv: 2610.00466 [astro-ph.IM].

[7] Bull, Philip, El-Makadema, Ahmed, ... **Wilensky**, **Michael J.**, et al. "RHINO: a large horn antenna for detecting the 21 cm global signal". In: *RAS Techniques and Instruments* 4, rzaf046 (Jan. 2025), rzaf046. DOI: 10.1093/rasti/rzaf046.

[8] Chen, Kai-Feng, **Wilensky**, **Michael J.**, Liu, Adrian, et al. "Impacts and Statistical Mitigation of Missing Data on the 21 cm Power Spectrum: A Case Study with the Hydrogen Epoch of Reionization Array". In: *Astrophysical Journal* 979.2, 191 (2025). DOI: 10.3847/1538-4357/ad9b91.

[9] Hughes, Thomas, **Wilensky**, **Michael J.**, and Bull, Philip. "Systematic assessment of the Hubble tension via Bayesian jackknife testing". In: *arXiv e-prints*, arXiv:2511.19341 (Nov. 2025), arXiv:2511.19341. DOI: 10.48550/arXiv.2511.19341. arXiv: 2511.19341 [astro-ph.CO].

[10] Kittiwisit, Piyanat, Murray, Steven G., ... **Wilensky**, **Michael J.**, et al. "matvis: a matrix-based visibility simulator for fast forward modelling of many-element 21 cm arrays". In: *RAS Techniques and Instruments* 4, rzaf001 (Jan. 2025), rzaf001. DOI: 10.1093/rasti/rzaf001.

[11] Pascua, Robert, Martinot, Zachary E., ... **Wilensky**, **Michael J.**, et al. "A Generalized Method for Characterizing 21 cm Power Spectrum Signal Loss from Temporal Filtering of Drift-scanning Visibilities". In: *Astrophysical Journal* 985.1, 127 (May 2025), p. 127. DOI: 10.3847/1538-4357/adc37d.

[12] Rath, E., Pascua, R., ... **Wilensky**, **Michael J.**, et al. "Investigating mutual coupling in the hydrogen epoch of reionization array and mitigating its effects on the 21-cm power spectrum". In: *Monthly Notices of the Royal Astronomical Society* 541.2 (Aug. 2025), pp. 1125–1144. DOI: 10.1093/mnras/staf1012.

[13] **Wilensky**, **Michael J.**, Morales, Miguel F., Hazelton, Bryna J., et al. "Erratum: "Evidence of Ultrafaint Radio Frequency Interference in Deep 21 cm Epoch of Reionization Power Spectra with the Murchison Wide-field Array" (2023, ApJ, 957, 78)". In: *Astrophysical Journal* 983.2, 187 (Apr. 2025), p. 187. DOI: 10.3847/1538-4357/adc813.

[14] **Wilensky, Michael J.**, Bull, Philip, and Fagnoni, Nicolas. "High dimensional beam inference II: inference of a perturbed HERA beam from simulated visibility data". In: *RAS Techniques and Instruments* 4, rzaf042 (Jan. 2025), rzaf042. DOI: 10.1093/rasti/rzaf042.

[15] **Wilensky, Michael J.**, Irfan, Melis O., and Bull, Philip. "Bayesian evidence for flux scale errors in Galactic synchrotron maps". In: *Monthly Notices of the Royal Astronomical Society* 539.4 (June 2025), pp. 3122–3137. DOI: 10.1093/mnras/staf668.

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[18] Garsden, Hugh, Bull, Philip, **Wilensky**, **Michael J.**, et al. "A demonstration of the effect of fringe-rate filtering in the hydrogen epoch of reionization array delay power spectrum pipeline". In: *Monthly Notices of the Royal Astronomical Society* 535.4 (2024). DOI: 10.1093/mnras/stae2541.

[19] Glasscock, Katrine A., Bull, Philip, and ... **Wilensky**, **Michael J.** "Statistical estimation of full-sky radio maps from 21 cm array visibility data using Gaussian constrained realizations". In: *RAS Techniques and Instruments* 3.1 (2024). DOI: 10.1093/rasti/rzae041.

[20] **Wilensky**, **Michael J.**, Burba, Jacob, Bull, Philip, et al. "High-dimensional inference of radio interferometer beam patterns I: parametric model of the HERA beams". In: *RAS Techniques and Instruments* 3.1 (2024). DOI: 10.1093/rasti/rzae029.

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[23] **Wilensky**, **Michael J.**, Brown, Jordan, and Hazelton, Bryna J. "Why and when to expect Gaussian error distributions in epoch of reionization 21-cm power spectrum measurements". In: *Monthly Notices of the Royal Astronomical Society* 521.4 (2023). DOI: 10.1093/mnras/stad863.

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[31] **Wilensky**, **Michael**. "Improving 21-cm Epoch of Reionization Power Spectrum Limits by Characterizing and Mitigating Radio Frequency Interference". PhD thesis. University of Washington, Seattle, Jan. 2021.

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[35] Zhang, Zheng, Pober, Jonathan C., ... **Wilensky**, **Michael J.**, et al. "The impact of tandem redundant/sky-based calibration in MWA Phase II data analysis". In: *Publications of the Astronomical Society of Australia* 37, e045 (2020). DOI: 10.1017/pasa.2020.37.

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[37] Li, W., Pober, J. C., ... **Wilensky**, **Michael J.**, et al. "First Season MWA Phase II Epoch of Reionization Power Spectrum Results at Redshift 7". In: *Astrophysical Journal* 887.2, 141 (2019). DOI: 10.3847/1538-4357/ab55e4.

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[40] **Wilensky**, **Michael J.**, Morales, Miguel F., Hazelton, Bryna J., et al. "Absolving the SSINS of Precision Interferometric Radio Data: A New Technique for Mitigating Faint Radio Frequency Interference". In: *Publications of the Astronomical Society of the Pacific* 131.1005 (2019). DOI: 10.1088/1538-3873/ab3cad.

Mentorship/Teaching/Outreach

1. Mentoring undergraduate at McGill University to extend SSINS technique from paper [40] to a Bayesian analysis of RFI properties in MWA data from paper [25]. This functions as a pilot study for the next entry in this list.

2. Mentoring a PhD student from Brown University in hierarchical Bayesian analysis of RFI from data in [25] based on Hidden Markov models. Will serve to generate automated seasonal statistics of RFI in radio telescope data. Assisted by previous entry in this list.

3. Thesis advisor for an M.Sc. student at University of Manchester investigating the Hubble tension via Bayesian hierarchical modeling. Publication in preparation.

4. Guided an investigation into using gaussian constrained realizations for inpainting flagged data in 21-cm power spectrum estimates. Consists mainly of informal discussions on the topic, usually in the form of providing technical feedback for problem-solving obstacles. **We co-authored a paper** [22].

5. Designed and guided a mathematical investigation into precise propagation of error bars through 21-cm power spectrum estimation pipeline with an undergraduate researcher. Consisted of weekly meetings to discuss mathematical concepts, software debugging, and general research techniques. **We co-authored a paper** [23].

6. Taught a statistics module for PhD level students at Queen Mary University of London. Duties mainly consisted of constructing guided tutorials through various statistical concepts using Python. Course material spanned basic Bernoulli urn style problems all the way to generalized linear regression/Wiener filtering.

7. Advised an undergraduate student through analysis and identification of subtle instrumental effects through sensitive quality metrics. Also advising them through enhancement of RFI filtering algorithm in SSINS software.

8. Introduced CHAMP student to fundamental radio astronomy concepts, python programming, and high performance computation. Advised student through analysis of HERA radio frequency interference environment using SSINS software.

9. Similar to previous entry, but not through CHAMP program.

10. Hosted one student through University of Washington Out in STEM (OSTEM) shadowing program.

Talks

1. URSI NRSM, 2025, "Evidence for uncorrected gain factors in Galactic synchrotron template maps," CU Boulder

2. *Invited talk (informal)*, PanEx journal club, 2024, "Evidence for uncorrected gain factors in Galactic synchrotron template maps," (delivered remotely)

3. *Invited talk*, CITA Jamboree, 2024, "A small survey of systematic challenges in 21-cm cosmology," CITA

4. *Invited talk (informal)*, LOFAR EoR Review Club, 2024, "Towards a constrainable analytic model of the HERA beams," ASTRON (delivered remotely)

5. *Invited panel member*, 21cmWG Precision Calibration Workshop, 2024, RFI Panel member, McGill University

6. LIM Workshop, 2024, "Chiborg: a Bayesian jackknife framework for testing consistency of multiple measurements," University of Illinois Urbana-Champaign

7. SALF IX, 2023, "Evidence of Ultra-faint RFI in Deep 21-cm Power Spectra," ASTRON
8. *Invited Talk*, Trottier Space Institute Cosmo-ph discussion, 2023, "Evidence of Ultra-faint RFI in Deep 21-cm Power Spectra with the Murchison Widefield Array," McGill University
9. *Invited Talk*, 2023, "Chiborg: a Bayesian jackknife framework for testing consistency of multiple measurements," Center for Computational Astrophysics
10. *Invited Seminar*, Yale Wright Lab Seminar, 2023, "Addressing the Challenges in 21-cm Cosmology using Bayesian Inference and other Data Analysis Techniques," Yale University
11. YERAC, 2023, "Chiborg: a Bayesian jackknife framework for testing consistency of multiple measurements," University of Manchester
12. URSI GASS, 2023, "Chiborg: a Bayesian jackknife framework for testing consistency of multiple measurements," Sapporo, Japan
13. MWA Project Meeting, 2023, "Evidence of Ultra-Faint RFI in Deep 21-cm Power Spectra," 2023, Curtin University, **won best talk**
14. *Invited Seminar*, Cambridge Summer Intern Seminar, 2023, "Mapping out the Early Universe by Observing Cosmic Radio Waves," University of Cambridge
15. SKA Cosmology Science Working Group Meeting, 2023, "Chiborg, a Bayesian jackknife framework for testing consistency of multiple measurements," University of Manchester
16. RFI2022, 2022, "Predicting and Measuring the Effect of RFI on 21-cm Epoch of Reionization Power Spectrum Measurements," Virtual
17. *Invited Seminar*, 2021,"Improving 21-cm Epoch of Reionization Power Spectrum Limits by Characterizing and Mitigating Radio Frequency Interference," Queen Mary University of London
18. URSI, "Consequences of RFI and RFI Excision for 21-cm EoR Science," 2020, Virtual
19. SALF VI, 2019, "Characterizing RFI Contamination in Epoch of Reionization Power Spectra," Arizona State University
20. *Invited Colloquium* ASTRO3D Visitor Colloquium, 2019, "Characterizing and Mitigating Radio Frequency Interference in Reionization Cosmology," University of Melbourne
21. MWA Project Meeting, 2019, "The SSINS of the MWA EoR Highband," Brown University
22. URSI, 2019, "A New Technique for Ultra-Faint RFI Contamination," University of Colorado at Boulder
23. MWA Project Meeting, 2018, "A New Technique for Ultra-Faint RFI Contamination," Nagoya University

Service

1. Monthly Notices of the Royal Astronomical Society Referee
2. Astrophysical Journal Referee
3. Astrophysics and Space Science Referee
4. Trottier Space Institute Undergraduate Research Fellowship Commitee
5. NASA ADAP Referee
6. Postdoc hiring committee for HERA research group at JBCA

Software

1. Sky-Subtracted Incoherent Noise Spectra (SSINS): https://github.com/mwilensky768/SSINS
2. Chiborg: https://github.com/mwilensky768/Chiborg